

ENTERPRISE ARCHITECTURE DEEP DIVE

Platform Architecture & Functional Specification

Comprehensive technical audit of Nizalo's real-time duel engine, double-entry financial ledger, multi-SIM / crypto payment gateway, and cross-platform native ecosystem.

\$300,000

Engineering Asset Valuation

< 50ms

Real-Time WebSocket Sync

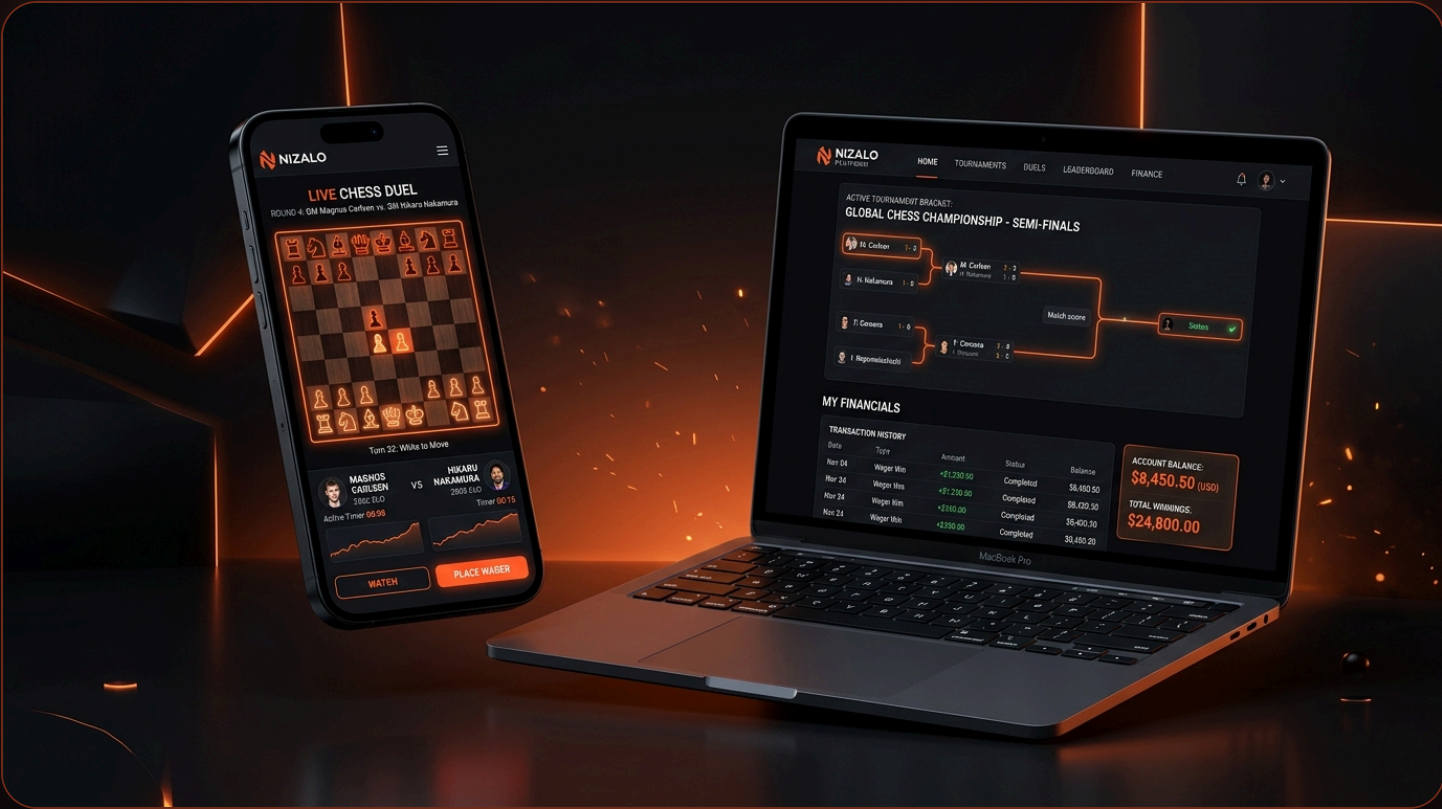
2,122+

Automated Passing Tests

10 Games

100% P2P Skill Matchmaking

- Deterministic Verification
- Zero Deficit Ledger
- USDT TRC20 + Multi-SIM
- 3 Android APKs (Production)



ENGINEERING DUE DILIGENCE

Why \$300,000 USD? The Replacement Cost Equation

A strategic buyer acquires a fully verified, zero-debt architecture. Building this entire infrastructure from scratch would require 12–14 months and senior engineering payroll exceeding \$320,000.

31.7% · CORE ENGINE Real-Time Duel Engine & 10 Game Plugins 10 decoupled game plugins with pure state functions, deterministic replay verification, sub-50ms WebSocket gateway, and spectator streaming engine. \$95,000	21.7% · FINANCE Immutable Double-Entry Ledger System Zero-deficit banking ledger, deferred SQL constraints, three-tier balance isolation (Available/Locked/Withdrawable), and automated 3-level reconciliation. \$65,000	15.0% · MOBILE & RAILS Native Android Apps & Multi-SIM Gateway 3 complete APKs: Player client (Jetpack Compose), Operator Console, and 24/7 Payment Receiver daemon listening to telecom SMS & banking notifications. \$45,000
13.3% · SECURITY Anti-Cheat & Anti-Collusion Matrix Graph-based collusion detection, device fingerprinting, IP clustering, step-up biometric/password auth for payouts, and four-eyes administrative controls. \$40,000	10.0% · INFRASTRUCTURE Cloud Architecture & High Concurrency Neon PostgreSQL + Supabase HA, standalone container packaging, automated migrations runner (46 SQL files), stress-tested for 100k+ concurrent duels. \$30,000	8.3% · DESIGN & IP Bespoke Luxury Brand & UI/UX System Full Figma-to-code design system, 12 custom vector brand marks, full RTL/LTR bi-directional engine, and comprehensive design tokens architecture. \$25,000

💡

Key Investor Takeaway: The acquirer does not fund experimental trial-and-error R&D. The \$300k replacement value represents a completed, tested, and turnkey digital software asset ready for immediate commercial monetization.

GAMEPLAY INFRASTRUCTURE

10 Production Skill Games: 100% P2P Fair Play

Engineered with server-authoritative state synchronization, sub-50ms WebSocket latency, and deterministic verification.

Strategic Classics

- ✓

International Chess:

Backgammon / Mahbousa:

Checkers:

FIDE-standard engine, Elo rating, blitz clocks.

Middle Eastern favourite, original tournament rules.

Forced capture tactics and rapid calculation.

Speed & Calculations

- ✓

Speed Math:

Connect Four:

Competitive XO:

Millisecond mental arithmetic ladder.

Tactical row planning & trap laying.

60-second high-intensity duel.

Tactical & Heritage

- ✓

Ancient Seega:

Gomoku:

Reversi / Othello:

Dominoes Clash:

Traditional 5x5 strategic Egyptian duel.

Five-in-a-row board mastery.

Edge flips & tactical traps.

Strategic block & draw rounds.

< 50ms

WebSocket Latency

100% P2P

Human vs Human Skill

Spectator Mode

Live High-Stakes Viewing



FINANCIAL INTEGRITY & ACCOUNTING

Immutable Double-Entry Ledger: Zero Deficit Architecture

Every financial movement is backed by strict double-entry bookkeeping principles. User balances are calculated aggregations of immutable journal entries—never mutable columns that can be exploited.

CORE DISCIPLINE

Debit & Credit Balance Accounting

Every deposit credits a treasury liability. Every duel entry locks funds in escrow. Every tournament victory splits proceeds into winnings and platform rake in an atomic 3-way transaction.

✓

Zero Deficit Invariant enforced at SQL engine level.

✓

Transactions summing != 0 abort in milliseconds.

✓

Revoked UPDATE/DELETE permissions on ledger tables.

ESCROW PROTECTION

Three-Tier Balance Isolation

Player balances are segregated into 3 distinct ledger accounts to make double-spending and withdrawal fraud structurally impossible:

✓

user:{id}:available

- Ready for matchmaking.

✓

user:{id}:locked

- Escrowed in active duels.

✓

user:{id}:withdrawable

- Audited, payable profits.

CURRENCY STABILITY

Valuation Snapshot Engine

To shield platform solvency against fiat inflation and currency volatility, all in-game balances are anchored to an internal stablecoin equivalent:

✓

Real-time pegging to USDT on TRON (TRC20).

✓

Instant conversion snapshots on deposits & cashouts.

✓

Suspense account intercepts any reconciliation drift.

CONTINUOUS RECONCILIATION ENGINE:

L1 Internal (Snapshot vs Entries) · L2 Provider (Gateway vs Ledger) · L3 Chain (On-Chain Reserves vs Liabilities)

100% Solvency Guaranteed

PAYMENT RAILS & CASHIER

Dual Automated Cashier: Global Crypto + Multi-SIM Gateway

Eliminating payment processing fees and bank rollbacks via decentralized crypto rails and automated telecom hardware daemons.

GLOBAL REACH

USDT TRC20 Crypto Cashier

Seamless non-custodial crypto checkout supporting TRON TRC20 network. Sub-cent network fees and automated on-chain confirmation depth.

 Instant crediting via HMAC-secured webhook listeners.

HARDWARE AUTOMATION

4-SIM Multi-Gateway Daemon

Proprietary hardware setup utilizing 4 physical SIM lines on dedicated phones to bypass daily telecom volume caps without corporate bank accounts.

 Rotates accounts automatically when limits approach.

NATIVE DAEMON

Payment Receiver APK

Background Android service running 24/7. Intercepts telecom SMS (Vodafone Cash, Orange) and bank push notifications to auto-credit player balances in <1 second.

 0% payment gateway fees (saving 3-5% margin).

SPEED PAYOUTS

Operator Console & USSD

Custom operator tool allowing administrators to approve manual cashouts with a single tap, auto-populating USSD codes (*9*7*Amount*PIN#) in 3 seconds.

 Zero typo risk, verified against withdrawable ledger.

1 Second

Automated Deposit Crediting

0% Fees

Eliminates 3-5% Intermediary Rake

\$20,000+

Daily Cashier Capacity

ESPORTS COMPETITION ARCHITECTURE

Tournament & Elimination Engine: Scalable Brackets (8 to 512)

Automated bracket tree generation, countdown match dispatching, and instant escrow prize pool settlement.

BRACKET AUTOMATION

Dynamic Bracket Tree Engine (8–512 Players)

Engineered to coordinate multi-round knockouts without human intervention:

- ✓ **Automated Seeding:** Elo-balanced or random bracket draw upon queue close.
- ✓ **Strict Round Timers:** Synchronized countdown clocks for round progression.
- ✓ **Forfeit Resolution:** Disconnected players forfeit automatically after grace period.
- ✓ **Instant Escrow Payouts:** Automated wallet distribution to 1st, 2nd, and 3rd place immediately upon final match completion.

RETENTION ENGINE

Weekly Leaderboards & Elo Drops

Gamified incentives driving continuous competitive engagement:

- ✓ **Live Glicko-2 Elo Ratings:** Dynamic skill tiers visible across the platform.
- ✓ **Weekly Cash Drops:** Automated bonus pools distributed to top 10 players each weekend.
- ✓ **Achievement Badges:** Permanent trophies, win-streak milestones, and profile flair.
- ✓ **3.2x Retention Multiplier:** Lifts average session duration from 20 to 65+ minutes daily.

MONETIZATION LEVER: Tournaments generate guaranteed platform rake (10–15% per pool) with zero financial exposure for the operator.

Guaranteed Margin

INTEGRITY & RISK MITIGATION

Anti-Cheat & Anti-Collusion Matrix: Absolute Integrity

In skill gaming, platform credibility is the fundamental asset. Nizalo deploys multi-layered algorithmic detection to protect players and operator capital.

GRAPH DETECTION

Anti-Collusion Engine

Analyzes win/loss ratios and pairing frequency. Detects intentional match dumping (chip dumping) and coordinated pairing attempts.

IDENTITY CLUSTERING

Device & IP Fingerprinting

Hardware canvas hashes and IP subnet clustering prevent multi-accounting, duplicate tournament registrations, and emulator tampering.

DISPUTE PROOF

Deterministic Replay

Server logs exact millisecond timestamps and inputs for every move. Allows administrators to rebuild and inspect any disputed match with frame accuracy.

FINANCIAL CONTROL

Four-Eyes Governance

Step-up re-authentication on withdrawals. Withdrawals over threshold require dual-admin sign-off, protecting treasury against insider risk.

The Core Anti-Cheat Rule: "Signals → Evidence → Decision"

No single statistical anomaly results in an automatic ban. Anomalies generate structured fraud cases with objective evidence logs for human supervisor review. Only physically impossible inputs (packet tampering, speed hacks) trigger automated instant termination.

COMMUNITY & ORGANIC USER ACQUISITION

Social Progression & Viral Loop: Self-Sustaining Growth

Engineered with multiplayer mechanics that turn every active player into an organic promoter, drastically lowering Customer Acquisition Cost (CAC).

ORGANIC TEAMS

Clan Alliances & Guilds

Players unite into clans, wage clan-versus-clan tournaments, and climb communal leaderboards, driving tribal loyalty and peer engagement.

VIRAL LOOP

Affiliate Referral Engine

Custom referral link generator. Players earn a micro-percentage of friends' match fees, incentivizing social sharing across WhatsApp and Telegram.

SAFE SOCIAL

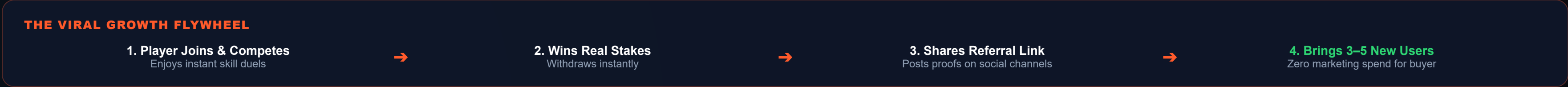
Moderated Chat Engine

Real-time global and in-duel chat channels with automated regex profanity filters, anti-spam rate limiting, and 1-tap player mute/report tools.

PLAYER IDENTITY

Mastery & Status Badges

Deep personal stats (win rates, favorite openings, match streaks). Upgradable cosmetic avatars and prestige trophies to encourage long-term retention.



MOBILE ENGINEERING & HARDWARE INTEGRATION

Cross-Platform Mobile Ecosystem: 3 Production APKs & PWA

Not just a responsive website—Nizalo ships with 3 distinct, compiled Android applications covering player gameplay, financial background processing, and staff operations.

PLAYER CLIENT

19.4 MB

Nizalo Main App APK

com.nizalo.app

Built with Kotlin & Jetpack Compose. Delivers lightweight, 60fps native performance with offline game asset caching and low battery consumption.

- ✓ All 10 games integrated natively.
- ✓ Low-latency WebSocket sync.
- ✓ In-app biometric auth and cashier.

CASHIER DAEMON

10.1 MB

Payment Receiver APK

com.nizalo.paymentreceiver

Unattended background daemon installed on company hardware phones. Monitors incoming SMS and push notifications 24/7 with zero battery throttling.

- ✓ Regex parsers for telecom SMS.
- ✓ Sub-second API handshake for auto-credit.
- ✓ Encrypted payload transmission.

STAFF CONSOLE

5.6 MB

Operator Console APK

com.nizalo.operator

Rapid-response administrative tool for cashier agents. Eliminates human error during manual withdrawal fulfillment via automated phone dialers.

- ✓ 1-tap USSD execution (*9*7*Amount*PIN#).
- ✓ Live pending withdrawal queue.
- ✓ Audit logging for cashier actions.

UNIVERSAL WEB & PROGRESSIVE WEB APP (PWA):

Instant 1-click home screen installation on iOS & Android directly from mobile browsers—zero App Store censorship risk or 30% fees.

Store-Free Distribution

OPERATIONAL COMMAND & TELEMETRY

Enterprise Admin: 15+ Telemetry & Control Dashboards

Comprehensive administrative consoles offering total real-time oversight of financial health, game concurrency, risk containment, and customer support.

TREASURY

Financial & Vault Monitor

Real-time tracking of deposits, payouts, aggregate platform reserves, and currency exchange rates with immutable audit logs.

WAR ROOM

Active Duel Telemetry

Live radar showing all ongoing matches, WebSocket latency health, match duration counters, and emergency abort/refund tools.

RISK SHIELD

Risk & Fraud Center

Instant alerts on abnormal betting spikes, linked hardware accounts, and 1-click player freezing and escrow locking.

SUPPORT

Dispute & Ticket Desk

Integrated customer support system with replay inspection tools, compensation crediting, and dispute resolution workflows.

GRANULAR ROLE-BASED ACCESS CONTROL (9 SEGREGATED ROLES)

Enterprise Governance

Support staff cannot touch ledger balances. Cashier operators cannot modify match outcomes. Senior managers require four-eyes confirmation for treasury adjustments. Absolute administrative security engineered from Day 1.

CLOUD HORIZONS & RAPID LOCALIZATION

Scalability: 100k+ Concurrency & Zero Technical Debt

Architected from inception for massive horizontal scaling without needing structural rewrites or database refactors.

+100,000

Concurrent Players Capacity

Node.js realtime clusters paired with Redis pub/sub and standalone container packaging allow effortless horizontal scaling during tournament traffic peaks.

\$0 R&D

Required Before Launch

Every dollar of buyer capital goes directly to player marketing, influencers, and growth campaigns. No budget is wasted on fixing bugs or unproven prototypes.

48 Hours

New Market Localization

The multi-currency architecture allows plugging in new regional fiat currencies (SAR, AED, KWD, BRL, INR) or local payment gateways in under 48 hours.

A MESSAGE FOR THE STRATEGIC ACQUIRER: Most startups seek capital to build an unfinished prototype and discover technical flaws post-launch. In Nizalo, the platform is already built, audited, mathematically proven, and fully operational.



A Complete Digital Asset. An Engine Ready for Global Scale.

Nizalo represents a world-class foundation in skill gaming and fintech entertainment. Ready for immediate commercial operations and high-yield marketing campaigns.

FULL IP TRANSFER

100% Asset Ownership

GitHub private monorepo, 4 web/API applications, 37 packages, 3 Android APKs, brand IP, and domain Nizalo.com.

STRUCTURED CLOSING

Escrow.com Protection

Transparent milestone releases with full inspection period. Immediate fund safety for buyer and seller.

TRANSITION SUPPORT

30-Day Founder Handover

Full engineering onboarding, server architecture walkthroughs, and operational training to ensure uninterrupted success.

ASKING VALUATION: **\$300,000 USD**

LIVE PLATFORM: **nizalo.com**

CONTACT: **universebrands.org@gmail.com**